

DNS 매뉴얼

<pre>[root@localhost conf.d]# dnf -y install bind bind-utils Last metadata expiration check: 0:10:55 ago on Wed Oct 22 15:21:46 2025. Package bind-utils-32:9.16.23-31.el9_6.x86_64 is already installed. Dependencies resolved. ===== Package Arch Version Repository Size ===== Installing: bind x86_64 32:9.16.23-31.el9_6 appstream 488 k Installing dependencies: bind-dnssec-doc noarch 32:9.16.23-31.el9_6 appstream 45 k checkpolicy x86_64 3.6-1.el9 appstream 352 k policycoreutils-python-utils noarch 3.6-2.1.el9 appstream 71 k python3-audit x86_64 3.1.5-4.el9 appstream 83 k python3-bind noarch 32:9.16.23-31.el9_6 appstream 69 k python3-distro noarch 1.5.0-7.el9 appstream 36 k python3-libsemanage x86_64 3.6-5.el9_6 appstream 78 k python3-ply noarch 3.11-14.el9_0.1 baseos 103 k python3-policycoreutils noarch 3.6-2.1.el9 appstream 2.0 M python3-setools x86_64 4.4.4-1.el9 baseos 551 k =====</pre>	1. DNS 설치 설치 명령어: <code>dnf -y install bind bind-utils</code>
<pre>[root@localhost conf.d]# systemctl status named ○ named.service - Berkeley Internet Name Domain (DNS) Loaded: loaded (/usr/lib/systemd/system/named.service; disabled; preset: disabled) Active: inactive (dead) [root@localhost conf.d]#</pre>	2. 상태 확인 <code>[root@localhost conf.d]#</code> <code>systemctl status named</code> **설치명(bind)과 패키지명(named)이 다르다
<pre>[root@localhost named]# cd /etc [root@localhost etc]# ls named.* named.conf named.rfc1912.zones named.root.key [root@localhost etc]#</pre>	3. named가 포함된 파일 확인 **named.conf : DNS 메인 설정 파일 named.rfc1912.zones : 각 도메인의 설정 파일
<pre>// See /usr/share/doc/bind*/sample/ for example named configuration files // options { listen-on port 53 { any; }; listen-on-v6 port 53 { ::1; }; directory "/var/named"; dump-file "/var/named/data/cache_dump.db"; statistics-file "/var/named/data/named_stats.txt"; memstatistics-file "/var/named/data/named_mem_stats.txt"; secroots-file "/var/named/data/named.secroots"; recursing-file "/var/named/data/named.recursing"; allow-query { any; }; } /* * If you are building an AUTHORITATIVE DNS server, do NOT enable</pre>	4. named.conf 옵션 변경 → <code>vim named.conf</code> 11번째 줄 <code>options {listen-on port 53 { 127.0.0.1; };</code> <code>/ 127.0.0.1; → any; 로 바꿔준다.</code> <code>allow-query { localhost; };</code> <code>/ localhost; → any; 로 바꿔준다.</code>

<pre> type master; file "named.empty"; allow-update { none; }; }; zone "yw.com" IN { type master; file "yw.com.zone"; allow-update { none; }; }; </pre>	5. named.rfc1912.zones 옵션 변경 -> vim named.rfc1912.zones <pre> zone "yw.com" IN { type master; file "yw.com.zone"; allow-update { none; }; }; </pre> 본인이 정한 도메인으로 변경 후 붙여넣기
<pre> [root@localhost named]# cp -ap named.localhost yw.com.zone [root@localhost named]# ls data dynamic named.ca named.empty named.localhost named.loopback slaves yw.com.zone [root@localhost named]# </pre>	6. /var/named 디렉터리의 named.localhost를 복사하여 새 이름(yw.com.zone)으로 붙여넣기 -> cp -ap named.localhost yw.com.zone ls -> yw.com.zone 확인
<pre> \$TTL 10 @ IN SOA ns.yw.com. root.yw.com. (0 ; serial 10 ; refresh 1H ; retry 1W ; expire 3H) ; minimum NS ns.yw.com. A 172.16.27.153 AAAA ::1 ns IN A 172.16.27.153 www1 IN A 172.16.27.153 </pre>	7. yw.com.zone 옵션 변경 SOA ns.yw.com. root.yw.com 변경 *ns를 제외하고 원하는대로 설정 가능 NS ns.yw.com. * 뒤에 (.)을 잊지 않기 A 본인IP > ns IN A 172.16.27.153 * 이 형식에 맞춰 3 클래스를 원하는대로 설정하여 입력 가능

<pre>[root@localhost named]# named-checkzone yw.com /var/named/yw.com.zone zone yw.com/IN: loaded serial 0 OK [root@localhost named]#</pre>	8. zone 파일 상태 확인 → named-checkzone yw.com /var/named/yw.com.zone OK 나오면 정상
<pre>[root@localhost named]# systemctl enable --now named [root@localhost named]# systemctl status named ● named.service - Berkeley Internet Name Domain (DNS) Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: disabled) Active: active (running) since Wed 2025-10-22 16:27:04 KST; 1min 57s ago Main PID: 34388 (named) Tasks: 10 (limit: 22916) Memory: 25.9M CPU: 79ms CGroup: /system.slice/named.service └─34388 /usr/sbin/named -u named -c /etc/named.conf</pre>	9. named 서비스 활성화 → systemctl enable --now named : 지금 바로 실행 → systemctl status named
<pre>[root@localhost named]# cat /etc/resolv.conf # Generated by NetworkManager nameserver 8.8.8.8 [root@localhost named]# nslookup ns.yw.com 172.16.27.153 Server: 172.16.27.153 Address: 172.16.27.153#53 Name: ns.yw.com Address: 172.16.27.153 [root@localhost named]#</pre>	10. DNS 서버가 도메인을 제대로 해석하는지 테스트 → cat /etc/resolv.conf > dns 확인 → nslookup ns.yw.com 172.16.27.153 : 도메인을 172.16.27.153 서버에게 질의한 것 (나 자신에게 질의응답)

- Domain Based VirtualHosting

<pre><VirtualHost *:80> ServerName ns.yw.com DocumentRoot "/home/yw/" <Directory "/home/yw/"> AllowOverride None Options Indexes FollowSymLinks Require all granted </Directory> </VirtualHost> <VirtualHost *:80> ServerName ww1.yw.com DocumentRoot "/home/test/" <Directory "/home/test/"> AllowOverride None Options Indexes FollowSymLinks Require all granted </Directory> </VirtualHost> ~</pre>	1. vhost.zone 설정 → (경로) cd /etc/httpd/conf.d/ → vim vhost.conf 포트를 모두 :80 으로 변경 ServerName ns.yw.com 추가 기재 * 위에서 정한 도메인 이름으로 추가
<pre>[root@localhost conf.d]# cd /var/log/httpd [root@localhost httpd]# ls access_log access_log-20251014 error_log error_log-20251014 [root@localhost httpd]#</pre>	2. log 확인 access_log : httpd 서버가 정상 동작한 로그정보 error_log : 비정상 동작 로그
<pre>188 ErrorLog "logs/error_log" 189 190 # 191 # LogLevel: Control the number of messages logged to the error_log. 192 # Possible values include: debug, info, notice, warn, error, crit, 193 # alert, emerg. 194 # 195 LogLevel warn 196 222 # 223 CustomLog "logs/access_log" combined 224 </IfModule> 225</pre>	3. 관리를 위한 log 가져오기 (경로) cd /etc/httpd/conf → vim httpd.conf 188, 223번째 줄의 복사하여 아래와 같이 변경한다. ErrorLog"/var/log/httpd/ww1_error_log" CustomLog"/var/log/httpd/ww1_access_log" combined

<pre><VirtualHost *:80> ServerName ns.yw.com DocumentRoot "/home/yw" ErrorLog "/var/log/httpd/ns_error_log" CustomLog "/var/log/httpd/ns_access_log" combined <Directory "/home/yw/"> AllowOverride None Options Indexes FollowSymLinks Require all granted </Directory> </VirtualHost> <VirtualHost *:80> ServerName ww1.yw.com DocumentRoot "/home/test" ErrorLog "/var/log/httpd/ww1_error_log" CustomLog "/var/log/httpd/ww1_access_log" combined <Directory "/home/test/"> AllowOverride None Options Indexes FollowSymLinks Require all granted </Directory> </VirtualHost></pre>	4. vhost.zone 재설정 → (경로) cd /etc/httpd/conf.d/ → vim vhost.conf (ErrorLog"/var/log/httpd/ww1_error_log" " CustomLog"/var/log/httpd/ww1_access_log" combined) * 도메인 확인하여 기재하기 사진과 같이 기재 후 저장
<pre>[root@localhost conf.d]# systemctl reload httpd [root@localhost conf.d]# cd /var/log/httpd [root@localhost httpd]# ls access_log error_log ns_access_log ww1_access_log access_log-20251014 error_log-20251014 ns_error_log ww1_error_log [root@localhost httpd]#</pre>	5. 로그 파일 생성 확인 → systemctl reload httpd : reload 해주기 (경로)cd /var/log/httpd -> ls : 생성된 도메인별 에러 로그 확인

